

Aleksandar Avdalović

Ph.D. Student, Division of Computer Science and Engineering
Louisiana State University, Baton Rouge, LA, USA

Email: aavdal1@lsu.edu

Google Scholar: scholar.google.com/citations?user=zDij9NEAAAAJ

Website: avdalovic.github.io

Research Interests: Industrial control systems security; trustworthy machine learning; anomaly detection; conformal prediction; uncertainty quantification; cyber-physical systems; network security.

Education

Louisiana State University, Baton Rouge, LA, USA Aug. 2023 – Present

Ph.D. Student in Computer Science and Engineering

GPA: 3.855/4.000

Advisor: Prof. Elias Bou-Harb

Research focus: Trustworthy deep learning for industrial control system security.

University of Primorska, FAMNIT, Koper, Slovenia Oct. 2020 – Aug. 2023

M.Sc. in Data Science

Average grade: 9.07/10

University of Coimbra, Coimbra, Portugal Feb. 2022 – Jun. 2022

Master's thesis research visit

Thesis: *Translation of Random Forests into Loop-Free Imperative Programs for Formal Verification*

Advisor: Prof. Raul Barbosa

Eötvös Loránd University (ELTE), Budapest, Hungary Feb. 2021 – Jun. 2021

ERASMUS+ Exchange Student, Applied Mathematics

University of Primorska, FAMNIT, Koper, Slovenia Oct. 2017 – Jul. 2020

B.Sc. in Mathematics

Average grade: 9.87/10

Publications

Peer-Reviewed Publications and Accepted Papers

1. **Aleksandar Avdalović**, Joseph Khoury, Ahmad F. Taha, and Elias Bou-Harb. *Enhancing Network Security Management in Water Systems using FM-based Attack Attribution*. IEEE/IFIP Network Operations and Management Symposium (NOMS), 2025. [doi] [arXiv]
2. **Aleksandar Avdalović**, Joseph Khoury, and Elias Bou-Harb. *GRACE-ICS: Graph-Aware Conformal Intervals for Industrial Operations Assurance*. IEEE/IFIP Network Operations and Management Symposium (NOMS), 2026. Accepted / to appear.
3. Anthony Nasry Massaad*, **Aleksandar Avdalović***, Peyton Andras, Joseph Khoury, and Elias Bou-Harb. *Beyond Arbitrary Thresholds: Conformal Prediction for Trustworthy CAN Bus Intrusion Detection*. IEEE International Conference on Communications (ICC), 2026. Accepted / to appear.
*Equal contribution.
4. Iat Tou Leong, **Aleksandar Avdalović**, and Raul Barbosa. *Program Verification for Rigorous Analysis of Decision Tree Ensembles Against Specifications*. IEEE/IFIP International Conference on Dependable Systems and Networks Workshops (DSN-W), 2025, pp. 55–62. [doi] [code]

Manuscripts Under Review

1. **Aleksandar Avdalović**, Joseph Khoury, Ibrahim AlJabea, Mustafa Hajij, and Elias Bou-Harb. *Beyond Flat Models: Preserving ICS Topology using Combinatorial Complexes for Hierarchy-Aware Intrusion Detection*. Manuscript under review.
2. **Aleksandar Avdalović**, Joseph Khoury, and Elias Bou-Harb. *Calibrating the Alarm: Why Deep Learning ICS Anomaly Detectors Fail at Deployment*. Manuscript under review.

Research Experience

Graduate Research Assistant, Cyber Threat Intelligence Lab

Aug. 2023 – Present

Louisiana State University, Baton Rouge, LA

Advisor: Prof. Elias Bou-Harb

- Develop methods for trustworthy deep learning intrusion detection in industrial control systems and cyber physical infrastructure.
- Designed intrusion attribution methods for water systems by modeling interactions between sensors and actuators with Factorization Machines.
- Developed conformal prediction methods that use system structure to provide calibrated uncertainty estimates for ICS forecasting.
- Built intrusion detection methods that preserve the Purdue hierarchy of ICS environments using combinatorial complexes and topological neural networks.
- Investigated deployment failures in deep learning ICS anomaly detectors and developed adaptive calibration methods for controlling false alarms.

Software and Systems Engineering Researcher

Feb. 2022 – Jun. 2022

Centre for Informatics and Systems, University of Coimbra, Portugal

- Studied how random forest models can be translated into loop free imperative programs for formal verification in safety critical settings.

Selected Research Projects

Deployment Realistic Calibration for ICS Anomaly Detection

- Showed that strong benchmark performance can still produce high false alarm rates in operation when thresholds are calibrated on historical benign data and transferred to new deployment regimes.
- Developed adaptive calibration methods based on conformal prediction that update thresholds online while keeping the upstream detector fixed.

Graph Aware Conformal Prediction for Industrial Operations Assurance

- Designed GRACE-ICS, a post-hoc and structure-aware conformal prediction layer that produces calibrated intervals for each sensor in ICS forecasting services.
- Evaluated the method on SWaT, WADI, and TEP using RNN, Transformer, and STGNN forecasters.

Hierarchy Aware Intrusion Detection for Industrial Control Systems

- Modeled ICS topology using combinatorial complexes to preserve devices, interactions, and PLC control zones.
- Developed anomaly scoring and localization methods that help operators identify where deviations arise in the control hierarchy.

Business Intelligence Consultant / Data Warehouse Engineer

Aug. 2022 – Aug. 2023

CRMT, Ljubljana, Slovenia

- Designed and implemented data warehouse and business intelligence solutions for organizational decision support.
- Worked with SQL, ETL pipelines, reporting systems, and analytics tools for data integration and visualization.
- Developed applied machine learning models for clients, including a case duration prediction prototype for a public sector client and a deployed next best offer recommender system integrated into an ETL pipeline for a telecommunications client.

Data Analyst Trainee

Jul. 2021 – Sep. 2021

ClearKarma, Vienna, Austria

- Supported market analysis, data extraction, data presentation, and prototype development for data-driven product features.

Honors and Awards

- Student Travel Grant, IEEE/IFIP Network Operations and Management Symposium (NOMS), 2025.
- University of Primorska Scholarship for Outstanding Achievements, 2017–2022.
- ERASMUS+ Scholarship for Master's Thesis Research Abroad, University of Coimbra, 2022.
- ERASMUS+ Traineeship Scholarship, Vienna, Austria, 2021.
- ERASMUS+ Exchange Scholarship, Eötvös Loránd University, Budapest, Hungary, 2021.
- Golden Medal, Primitijada Student Mathematical Competition, Golden Sands, Bulgaria, 2019.
- Competitor, Al-Khorezmi International Mathematical Olympiad, Urgench, Uzbekistan, 2018.

Languages

Serbian: Native

English: Full professional proficiency

Slovenian: Full professional proficiency